

SIMATS ENGINEERING

Artificial Intelligence (CSA17) — CO3 Assessment Tool 3

Questions and Answers

Q1. Which of the following best describes a Logical Agent in AI?

Answer: b) An agent that uses a knowledge base and logical inference to decide actions.

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

Answer: d) Both b and c.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

Answer: b) A conjunction of disjunctions (clauses).

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

Answer: b) Every entity that is a student passes.

Q5. Unification in FOL is the process of finding substitutions that:

Answer: b) Make two logical expressions syntactically identical.

Q6. In the Resolution algorithm, deriving the empty clause indicates:

Answer: c) A contradiction — the original goal is proven.

Q7. Forward Chaining is best described as a _____ strategy.

Answer: b) Data-driven / bottom-up.

Q8. Backward Chaining begins reasoning from:

Answer: b) The goal and works backward to verify supporting facts.

Q9. Which step is NOT part of converting a sentence to CNF?

Answer: c) Apply Skolemization to remove existential quantifiers.

Q10. Knowledge Engineering in FOL involves:

Answer: b) Defining domain concepts, relations, and axioms formally in FOL.

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

Answer: b) Q must be true.

Q12. The Most General Unifier (MGU) is the substitution that:

Answer: b) Makes two terms identical with minimal variable binding.

Q13. Which inference method is more appropriate when a specific goal is known?

Answer: c) Backward Chaining.

Q14. Propositional Logic differs from First Order Logic because it:

Answer: b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions.

Q15. Resolution Refutation proves a goal G by:

Answer: b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause).

Q16. Learning from observation in AI refers to:

Answer: b) Improving performance by examining examples or experience from the environment.

Q17. Inductive learning is best described as:

Answer: b) Generalizing a hypothesis from specific training examples.

Q18. In a Decision Tree, what does each internal node represent?

Answer: c) A test on an attribute / feature.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

Answer: c) Uses prior domain knowledge to explain and generalize a single example.

Q20. In statistical learning, a Naive Bayes classifier assumes:

Answer: c) All features are conditionally independent given the class label.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

Answer: c) Directly computing maximum likelihood estimates from observed data.

Q22. The Expectation-Maximization (EM) algorithm is used when:

Answer: c) The dataset contains hidden (latent) variables that cannot be directly observed.

Q23. In a Neural Network, the activation function serves to:

Answer: c) Introduce non-linearity to enable learning of complex patterns.

Q24. Which of the following correctly describes backpropagation in a neural network?

Answer: c) The error is propagated backward from output to input layers to adjust weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

Answer: c) Finding the maximum-margin hyperplane that separates classes.

Q26. The kernel trick in kernel machines (SVMs) allows:

Answer: c) Operating in a high-dimensional feature space without explicitly computing coordinates.

Q27. In Reinforcement Learning, the agent aims to maximize:

Answer: c) Cumulative reward received over time through interaction with the environment.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

Answer: c) A model-free, off-policy temporal difference learning method.

Q29. Overfitting in machine learning occurs when:

Answer: c) The model fits training data too closely and generalizes poorly to new data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

Answer: c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples.